Conditional Bounds for Representation-Based Subset Sum via Profile Completeness and a Distinct-Sums Parameter

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***Abstract*—Whether subset sum can be solved in $2^{(1/2-\varepsilon)n}$ time on every input is open. The representation technique of Howgrave-Graham and Joux does far better on random inputs, but its worst-case analysis needs the representations of a solution to "mix" modulo a random prime. This note does *not* give a worst-case algorithm. It makes three precise statements about the technique with a balanced $2^{0.4057n}$ sub-solver. (1) Completeness of the sub-solver costs only a $\Theta(\sqrt{n})$ factor: a random permutation of the input balances a weight-$n/2$ solution with probability $\sqrt{8/(\pi n)}(1+o(1))$, so the $2^{0.811n}$ price of enumerating every weight profile is not needed. (2) Mixing for a single prime follows from one parameter: if the sums of the balanced half-weight subcollections of a solution's support take D^* distinct values and no input has too many exact sum-matching pairs, the expected time is $\mathrm{poly}(n)\big(2^{0.4057n}+(A+N_t)/D^*\big)$, below $2^{n/2}$ exactly when $D^* > 2^{0.311n}$. (3) When D^* is smaller, the support is forced to contain an equal-sum pair of subsets of length at most k, where k/n is explicit (0.105 at the threshold), and *if* such a relation lies in the solution the remaining cost is at most $2^{0.492n}$ whenever $D^* \geq 2^{0.05n}$. The unproved step is bounding collisions among $\Theta(n)$-subsets of the whole input. We report measurements at $n \leq 40$ that agree with every proved statement and mark clearly what small sizes cannot show.**

***Index Terms*—subset sum, representation technique, meet-in-the-middle, additive energy, worst-case analysis**

I. Introduction

Given integers $a_1, \ldots, a_n$ and a target t, exact subset sum asks for $x \in \{0,1\}^n$ with $\sum_i a_i x_i = t$. Meet-in-the-middle [1] solves it in $\widetilde{\mathcal{O}}(2^{n/2})$ and nothing better is known on every input, apart from a polynomial saving [4]. On random ("hard knapsack") inputs the representation technique of Howgrave-Graham and Joux [2], improved by Becker, Coron and Joux [3], reaches $2^{0.337n}$ and $2^{0.291n}$. Randolph and Węgrzycki [5] recently beat meet-in-the-middle in the worst case for many coefficient sets, but not for $\{0,1\}$ (equivalently, Partition), which is the case studied here.

What this note contributes.: The technique has three places where an average-case argument must become a worst-case one: completeness of the sub-solver, mixing of the representations modulo a prime, and the exact matching step. We settle the first, reduce the second to a single combinatorial parameter, and reduce what is left to a collision-counting question. All proofs are given in Appendix A. Nothing here improves the worst-case exponent below $1/2$ unconditionally; the contribution is the sharp form of the remaining obstruction.

TABLE I
Balancing probability at $w = n/2$ (exact, 20 000 samples) and the permutations needed for 99% coverage.

n	exact q	$q\sqrt{n}$	sampled	perms
16	0.3807	1.523	0.3887	10
64	0.1971	1.577	0.1951	21
256	0.0994	1.591	0.1011	44
1024	0.0498	1.595	0.0510	91

Setting and notation.: Throughout $n \equiv 0 \pmod 8$, the target solution x has weight $n/2$ and support S, $|S| = n/2$. The sub-solver enumerates "balanced" subsets $y \subseteq [n]$ with $|y| = n/4$ and exactly $n/8$ elements in each of two fixed halves H_1, H_2 of the input; their number is $A = \binom{n/2}{n/8}^2 = 2^{0.811n}$, enumerated in $2\binom{n/2}{n/8} = 2^{0.4057n}$ time per list plus output (a balanced meet-in-the-middle on residue classes). The solution is written $x = y + z$ with y, z disjoint balanced; a modulus $p \approx \binom{n/4}{n/8}^2 \approx 2^{n/2}$ and a random residue r keep, in expectation, one representation. We treat weight $n/2$, the hardest case; other weights change constants.

II. Completeness is a polynomial factor

The sub-solver sees a representation only if the support is split $n/4$–$n/4$ across H_1, H_2. An earlier version of the companion note [8] treated this as a barrier and priced the repair at the cost of all weight profiles, $\sum_i \binom{n/2}{i}\binom{n/2}{n/4-i} = \binom{n}{n/4} = 2^{0.811n} > 2^{n/2}$. That price is unnecessary, because the input order is free.

Theorem 1 (Profile completeness)**.** *Let π be a uniformly random permutation of the inputs. A fixed support of size w (even) has exactly $w/2$ elements in each half with probability*

$$q_w = \binom{n/2}{w/2}^2 \Big/ \binom{n}{w}, \qquad q_{n/2} = \sqrt{\tfrac{8}{\pi n}}\,(1+o(1)).$$

Hence $\mathcal{O}(\sqrt{n}\log(1/\delta))$ permutations succeed with probability $1-\delta$, and an unknown weight adds a factor n (try each w). Table I compares the exact value, sampling, and the real search on planted solutions whose support lies entirely in one half: all were found within a budget of $4\sqrt{n}$ permutations at $n = 24, 32$.

Algorithm 1 Permuted representation search for a scale guess M

Require: weights $a_1, \dots, a_n$; target t; modulus scale M
Ensure: a subset $x \in \{0,1\}^n$ with $\sum_i a_i x_i = t$, or $\perp$

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draw a uniform permutation π of [n], defining halves H_1, H_2
draw a random prime p ∈ [M, 2M] and residue r ∈ {0, ..., p − 1}
Y_r ← {y ⊆ [n] : |y ∩ H_1| = |y ∩ H_2| = n/8, σ(y) ≡ r (mod p)}
Z_r ← {z ⊆ [n] : |z ∩ H_1| = |z ∩ H_2| = n/8, σ(z) ≡ t − r (mod p)}
for (y, z) ∈ Y_r × Z_r with σ(y) + σ(z) = t do ▷ hash on exact sum
    if y ∩ z = ∅ then
        return y ∪ z
    end if
end for
return ⊥
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III. A distinct-sums parameter controls a single prime

Mixing was previously controlled only on average over a window of primes, which by Markov gives a $1/\mathrm{polylog}$ fraction of good primes and loses the constant-fraction accounting needed for an exponent [8]. We show that one quantity suffices.

For a partition $S = S_1 \cup S_2$, $|S_i| = n/4$, let $T(S_1, S_2) = \{\sigma(y_1) + \sigma(y_2) : y_i \subseteq S_i,\ |y_i| = n/8\}$ and $D^* = \min |T(S_1, S_2)|$ over partitions. Let N_t be the expected (over the random permutation) number of pairs (y, z) of balanced $n/4$-subsets with $\sigma(y) + \sigma(z) = t$ (not necessarily disjoint), and $a_{\max} = \max a_i$. Algorithm 1 gives the permuted single-prime search.

Theorem 2 (Single-prime lift)**.** *Running Algorithm 1 over $O(n)$ guesses $M = 2^j$ finds a solution in expected time*

$$\mathrm{poly}(n, \log a_{\max})\Big(2^{0.4057n} + \frac{A + N_t}{D^*}\Big).$$

Corollary 1. *If $N_t \le A$ then the time is $\mathrm{poly}(n)\, 2^{0.4057n}$ for $D^* \ge 2^{0.4057n}$ and $\mathrm{poly}(n)\, 2^{0.811n}/D^*$ below it; this is below $2^{n/2}$ exactly when $D^* > 2^{0.311n}$.*

Remarks.: (i) Theorem 2 is of the same flavor as the Austrin–Kaski–Koivisto–Nederlof density dichotomy [7]: a large pair count N_t means large exact-sum bins and is their regime. We do not claim it is new, only that it is the clean form of the representation argument for one prime. (ii) Nothing in the proof assumes the inputs are random; only D^* and N_t enter. (iii) Random inputs have $D^* = \Theta(\binom{n/4}{n/8}^2) = 2^{n/2}/\mathrm{poly}$ and negligible N_t.

Table II agrees: the proved bound is conservative, and a low D^* (here 88) occurs only on arithmetic progressions, which small-doubling algorithms [6] solve anyway. A complementary measurement, $|Y_r|/\mathbb{E}|Y_r|$ for a random prime of the window at $n \le 40$, stayed within $1.3\times$ on every non-progression family (mean 1.00). Prime moduli matter: on the geometric family at $n = 32$ the default power-of-two modulus found the solution in only half the runs, a prime of the same size in all of them.

TABLE II
Coverage check (Theorem 2) at $n = 32$: balanced D^*, mean fraction of the D^* sums covered over the prime window, and the proved lower bound $1/(1 + D^* B/\pi_w)$. Planted supports, $Y = \binom{n/4}{n/8}^2 = 4900$.

family	D^*	mean cover	bound
random n-bit	4900	0.950	0.216
geometric	4900	0.952	0.227
two-scale	4900	0.950	0.118
gap rank 2	4748	0.956	0.327
q-multiples	4884	0.952	0.291
arithmetic prog.	88	1.000	0.261

IV. What a small D^* forces

By Corollary 1 the uncovered instances have $D^* \le 2^{0.311n}$, far fewer distinct sums than the $2^{n/2}$ representations. Pigeonhole forces collisions, and a counting argument bounds how long the colliding sets must be. Call S *k-dissociated* if distinct subsets of S of equal size $j \le k$ have distinct sums.

Lemma 1 (Forced relation)**.** *If S is $2j$-dissociated then $D^* \ge \binom{n/8+j}{j}^2$. Consequently, if $D^* < \binom{n/8+j}{j}^2$ there are disjoint $P, Q \subseteq S$ with $|P| = |Q| \le 2j$ and $\sigma(P) = \sigma(Q)$.*

The lemma need not be tight, and no construction is known that attains it. For a support whose coordinates are random (a box in $[0, L)^d$ with random generators) a birthday count puts the shortest relation at $k/n \approx h_2^{-1}(\delta)/2$ (about 0.029 at $\delta = 0.32$), still $\Theta(n)$ but far below the lemma's 0.11. Avoiding collisions as well as the Gilbert–Varshamov bound for codes gives $k/n \approx 0.041$ at the same δ. Reaching the lemma's value needs near-perfect B_k sets, which are not known.

Conditional use.: Write $D^* = 2^{\delta n}$ and $k = k(\delta)$ the smallest length allowed by the lemma. Finding an equal-sum pair of k-subsets among the n inputs by sorting costs $\binom{n}{k}$. *If* the pair found lies in S, the $\le 2k$ elements $P \cup Q$ are fixed in the solution and the remainder is solved by meet-in-the-middle in $2^{(n-2k)/2}$. The exponent is $\max\{h_2(k/n),\ (1 - 2k/n)/2\}$, listed with the Theorem 2 exponent in Table III. Below the crossover $\delta \approx 0.311$ the relation route is at most 0.492 for the values shown; above it Theorem 2 takes over. So *conditionally on the relation lying in the support* every instance with $D^* \ge 2^{0.05n}$ has exponent at most $0.492 < 1/2$; as $\delta \to 0$ the route tends to $1/2$, and tiny D^* is the natural domain of small-doubling algorithms [6] (not proved here).

A heuristic model.: Take the three relation-length models above (birthday, Gilbert–Varshamov, and the lemma's extremal value), random decoys (so that no relation shorter than $0.11n$ involves a decoy), and $|\Sigma(S)| \le 2^{\delta n}$ for the sums of all sizes. Running Theorem 2 and the relation route in parallel has exponent $\min\{\text{Thm. } 2, \max\{\text{find}, \text{compress}\}\}$ with compress

TABLE III
ASYMPTOTIC EXPONENTS (BASE 2, DIVIDED BY n). k/n: SMALLEST FORCED RELATION LENGTH. FIND: $\log_2 \binom{n}{k}/n$. RESIDUAL: $(1-2k/n)/2$. THM. 2: $\max\{0.4057, 0.811-\delta\}$.

δ	k/n	find	residual	route	Thm. 2
0.05	0.0078	0.066	0.492	0.492	0.761
0.20	0.0522	0.296	0.448	0.448	0.611
0.30	0.0992	0.466	0.401	0.466	0.511
0.311	0.1053	0.486	0.395	0.486	0.500
0.35	0.1290	0.555	0.371	0.555	0.461
0.40	0.1637	0.643	0.336	0.643	0.411

TABLE IV
RANK-d SUPPORTS PLUS DECOYS AT $n=32$: D (DISTINCT SUMS OF ALL WEIGHT-$n/4$ SUBSETS OF THE SUPPORT, SO $D^* \le D$), $Y = 12870$ SUCH SUBSETS, ADDITIVE-QUADRUPLE CORE SIZE, AND WHETHER IT CONTAINS THE SUPPORT.

d	D	D/Y	core	support in core
3	168	0.01	16	yes
4	522	0.04	16	yes
6	3260	0.25	16	yes
8	7642	0.59	4	no

$= 1/4 + \delta/2$. Its worst case over δ is 0.431 for the birthday and Gilbert–Varshamov models ($\delta \approx 0.38$) and 0.4913 for the extremal model ($\delta \approx 0.32$), always below $1/2$. This is a model, not a theorem: decoys are assumed random and $|\Sigma(S)|$ is not bounded by D^*.

The gap.: Nothing prevents a relation found among all n inputs from using decoys outside S. Let N_c be the number of equal-sum disjoint pairs of k-subsets among all inputs. If $N_c \le 2^{\epsilon n}$, branch on each (time $N_c\, 2^{(n-2k)/2}$); if N_c is large the input has many equal-sum subset pairs at scale $\Theta(n)$, which is the large-bin regime of [7], whose intermediate range is the open band. Bounding N_c against the bin size is the single missing lemma.

Small-size evidence.: Planted supports in a rank-d progression with random n-bit generators and random decoys (Table IV) have small D^*/Y, and for small rank the additive-quadruple core equals the support exactly: random decoys produced *no* spurious relations. At rank 8 ($n=32$) the core misses the support, i.e. relations are longer than 4 terms, which the lemma allows (it forces length only at scale $\Theta(n)$). A Sidon-like yet low-rank support needs n in the hundreds, so these tests probe only the small-rank corner.

V. DETECT AND COMPRESS (EXPERIMENTAL)

Everything in this section is measured or heuristic. For any sub-collection C of the input, a meet-in-the-middle with $\Sigma(C)+\Sigma(R_1)$ on one side and $\Sigma(R_2)$ on the other ($R = R_1 \cup R_2$ the rest) is exact and costs about $|\Sigma(C)| + \sqrt{2^{|R|}|\Sigma(C)|}$; it helps when C is compressible, $|\Sigma(C)| \ll 2^{|C|}$. This is the few-distinct-sums regime of [7], [6] applied to a sub-collection; we do not claim it is new. We choose C greedily: add the element that makes $\Sigma(C)$ grow least, preferring elements that occur in short equal-sum relations, and keep the best prefix. An element unrelated to C doubles $\Sigma(C)$, so decoys are left out wherever they sit.

Robustness and scale.: Decoys built to carry their own short relations ($a+b=c$ gadgets, elements linked to the support, and pairs $e, e+s_i$ that glue random elements into the support's relation component) defeat a solver that takes whole relation components, but not the greedy rule: at $n=32$ it is exact and 4.7–$12.2\times$ faster than meet-in-the-middle on every attack. With a rank-2 or rank-3 structured support it solves instances up to $n=180$ (verified witness, about $2.4\cdot10^7$ states against $1.2\cdot10^{27}$ for plain meet-in-the-middle), keeping 150 of the 154 chosen elements inside the support under the gluing attack. These supports have small doubling, so this shows the method scales, not that it reaches the hard band.

Where a hard instance could still hide.: The greedy rule sees structure only once a relation is complete, so the remaining question is whether a compressible support can have all its relations long. Pigeonhole bounds the shortest relation by s with $\sum_{j\le s}\binom{m}{j} > |\Sigma_{\le s}|$; random-like sets collide at the much earlier birthday scale, and algebraic B_h or MDS constructions give only $O(m/\log m)$. Searches up to $m=60$ found nothing beyond the birthday scale except a Sidon set at $m=40$ (one step). Under random decoys and $|\Sigma(S)| \le 2^{\delta n}$ for all sizes, a heuristic model puts the best of Theorem 2 and this route at exponent at most 0.431 (birthday model) or 0.4913 (pigeonhole-extremal relations). The second assumption follows, up to a factor n, from the unproved statement that $\max_k |\Sigma_k(X)| \le \mathrm{poly}(m)\,|\Sigma_{\lfloor m/2\rfloor}(X)|$ for every m-set X, for which all our searches found a ratio at most 1.032.

VI. LIMITS AND STATUS

Everything above is conditional or partial. Theorem 2 solves instances with $D^* > 2^{0.311n}$ and a controlled pair count in time between $2^{0.4057n}$ and $2^{n/2}$, never $2^{n/3}$: that needs a sub-solver faster than the balanced one. A three-level Becker–Coron–Joux evaluation at $n = 16, 32$ matches the predicted list sizes (leaf 1.00, middle 0.92–0.98, top 1.16–1.35 of the model) but asymptotics cannot be read from $n \le 32$, and total work including the $\Theta(\sqrt{n})$ permutations and polynomial factors exceeds meet-in-the-middle at these sizes: all measurements validate the structure, not a speedup. The discrete Cauchy–Schwarz bound, the disjoint forced-relation stripping, and the compressible-core merge are machine-checked in Lean; the probabilistic expectations and Stirling asymptotics are pen-and-paper.

Open problems.: (i) Prove or refute $\max_k |\Sigma_k(X)| \le \mathrm{poly}(m)\,|\Sigma_{\lfloor m/2\rfloor}(X)|$ for every set X of m integers. (ii) Is there a family of sets with $|\Sigma(X)| \le 2^{cm}$, $c<1$, whose shortest equal-sum relation is asymptotically longer than the birthday scale? A negative answer with a bound, together with (i) and a way to remove the random-decoy assumption, would turn the model of Section V into a theorem for every instance whose solution support has $D^* \le 2^{0.311n}$. The unconditional worst-case question below $2^{n/2}$ stays open.

APPENDIX A
PROOFS

A. Proof of Theorem 1

The first formula is the hypergeometric probability of splitting w chosen positions evenly across two halves of size $n/2$. By Stirling's approximation, $\binom{2a}{a} = 4^a/\sqrt{\pi a}\,(1+o(1))$; setting $a = n/4$ gives $\binom{n/2}{n/4}^2 = 2^n \cdot 4/(\pi n)$ and $\binom{n}{n/2} = 2^n/\sqrt{\pi n/2}$. Dividing the two yields $\sqrt{8/(\pi n)}\,(1+o(1))$. □

B. Proof of Theorem 2

Let π balance S (probability $q_{n/2}$, Theorem 1), giving the partition (S_1, S_2), and write $T = T(S_1, S_2)$ with $D = |T| \geq D^*$. Every balanced $y \subseteq S$ has balanced complement $S \setminus y$ in S. Thus if $r \equiv \sigma(y) \pmod p$ for some such y, the pair $(y, S \setminus y)$ lies in $Y_r \times Z_r$ (since Z_r has residue $t - r$).

Coverage. Let $c_\rho = \#\{s \in T : s \equiv \rho \pmod p\}$. Then $\sum_\rho c_\rho^2 = D + K_p$, where K_p is the number of ordered pairs $s \neq s'$ in T with $p \mid s - s'$. By Cauchy–Schwarz, at least $D^2/(D + K_p)$ residues are covered. Each nonzero difference has absolute value at most $na_{\max}$ and thus at most $B = \log(na_{\max})/\log M$ prime factors $\geq M$; averaged over the π_w primes of the window $[M, 2M]$, $\mathbb{E}_p K_p \leq D^2 B/\pi_w$. Since $u \mapsto D^2/(D+u)$ is convex on $u \geq 0$, Jensen's inequality gives $\mathbb{E}_p|T \bmod p| \geq D/(1 + DB/\pi_w)$. With $\pi_w \sim M/\ln M$ and $M \geq 2DB \ln M$, this expectation is at least $D/2$, so $\Pr_r[r \text{ covered}] \geq D/(4M)$.

Lists and matches. For every prime, $\sum_\rho |Y_\rho| = A$, so $\mathbb{E}_r|Y_r| = \mathbb{E}_r|Z_r| = A/p$. Each pair counted by N_t lies in exactly one $Y_r \times Z_{t-r}$ (its residue is fixed by y), so $\mathbb{E}_r[\#\text{exact pairs}] = N_t/p$. By Markov's inequality and a union bound with $\lambda = 24M/D$, the event "r covered, $|Y_r|, |Z_r| \leq \lambda A/p$, and exact pairs $\leq \lambda N_t/p$" holds with probability at least $D/(4M) - 3/\lambda = D/(8M)$.

Cost. Conditioning on a good attempt, the balanced split costs $2^{0.4057n}$ and the list construction and exact-sum matching cost at most $\lambda(A + N_t)/p \leq 48(A + N_t)/D$. The expected number of attempts is $\mathcal{O}(\sqrt{n}) \cdot 8M/D$, which is $\mathrm{poly}(n, \log a_{\max})$ for the scale guess $M \approx 2DB \ln M$. □

C. Proof of Lemma 1

Fix a partition (S_1, S_2) and choose $R_i \subseteq S_i$ with $|R_i| = n/8 - j$; set $U_i = S_i \setminus R_i$, so $|U_i| = n/8 + j$. For j-subsets $K_i \subseteq U_i$, the sum $\sigma(R_1 \cup R_2) + \sigma(K_1) + \sigma(K_2)$ lies in $T(S_1, S_2)$, and distinct pairs (K_1, K_2) yield distinct $2j$-subsets $K_1 \cup K_2$ of S, hence distinct sums by $2j$-dissociation. There are $\binom{n/8+j}{j}^2$ such pairs. Taking the contrapositive: if $D^* < \binom{n/8+j}{j}^2$, two distinct equal-size subsets of S of size at most $2j$ have equal sum; deleting their common elements leaves disjoint $P, Q \subseteq S$ with $|P| = |Q| \leq 2j$ and $\sigma(P) = \sigma(Q)$. □

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